

FINAL EXPLANATION: PHYSICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: PHYSICS GRADE 10: 4.2 INTRODUCTION TO SPACE PHYSICS

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation for the unit. Answer the driving question: How did everything we see in the night sky above Nairobi come to exist, and how do scientists gather evidence about a universe they cannot directly touch or travel to? Write your answers in the space under each prompt. Use evidence from the lessons (the Big Bang, the evolution of the universe, classifying celestial bodies, the Solar System, Kepler's Laws, gravity, telescopes and space missions). For each answer, give a claim, supporting evidence and your reasoning. Write in full sentences and use the correct science words. You may include labelled sketches if they help.

Part 1: The sky looks still, but is it?

Imagine you are standing in the Rift Valley or on the shore of Lake Victoria on a clear night. The thousands of stars, the Milky Way band, the Moon and planets such as Venus and Jupiter look permanent and unchanging. Scientists say none of this existed 13.8 billion years ago. (a) Explain why the sky looks unchanging to us even though science says it has a violent, dynamic history. (b) Describe how the Big Bang is different from an ordinary explosion inside space.

Write your answer here:

Part 2: From the Big Bang to the structured night sky

Put these events in order from earliest to latest: first stars, first atoms of hydrogen and helium, first galaxies, our Solar System, quark-gluon plasma, our Milky Way. Then explain how the universe changed from a hot, structureless state into the stars and galaxies visible from Nairobi tonight. In your explanation, describe what processes caused the change, and identify the type of galaxy the Milky Way is.

Write your answer here:

Part 3: Order in the Solar System and the force behind it

The planets Venus and Jupiter that we see above Nairobi belong to the Solar System. (a) Describe the layout of the Solar System, and explain what a scaled model of it (such as the corridor or toilet-paper model you walked) showed you about the space between the planets. (b) State Kepler's three laws in your own words. Then, using the orbital data you worked with for Mercury, Venus, Earth, Mars, Jupiter and Saturn, explain what you found when you compared the square of each planet's orbital period with the cube of its semi-major axis, and say what this told you about the planets. (c) Name the force that keeps the planets in orbit, state how it depends on mass and distance, and give one effect it has on Kenyan communities.

Write your answer here:

Part 4: Gathering evidence without going there

No human was present at the Big Bang and we cannot travel to distant galaxies. (a) Name at least three kinds of evidence or instruments scientists use and explain what each one tells us. (b) Explain why a telescope can be described as a time machine. (c) Give one example of how Kenya or Africa contributes to space science.

Write your answer here:

Part 5: Your final answer to the driving question

Write a final explanation of 6 to 8 sentences that answers the driving question: How did everything we see in the night sky above Nairobi come to exist, and how do scientists gather evidence about a universe they cannot directly touch or travel to? Use the Claim, Evidence, Reasoning structure. Include at least one idea about the universe's origin, one about the Solar System, and one about how evidence is gathered. Then say how your thinking has changed since Lesson 1.

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Scientific accuracy of the origin and evolution of the universe	Correctly describes the Big Bang as expansion of space from a singularity about 13.8 billion years ago, orders the cosmic events correctly and uses the given times accurately.	Describes the Big Bang and most events in the correct order, with minor errors in times or terms.	Describes the Big Bang as an ordinary explosion in space or places several events in the wrong order.
Use of evidence (CMB, expansion, Kepler's Laws, gravity, telescopes and missions)	Uses several specific pieces of evidence from the unit and explains clearly what each one shows.	Uses some evidence from the unit but explains what it shows only partly.	Gives few or no pieces of evidence, or only states facts without linking them to a claim.
Reasoning about indirect observation	Clearly explains that light travel time lets telescopes look back in time and that instruments extend human senses, and links this to the driving question.	Explains that instruments gather evidence from far away but is unclear about looking back in time.	Does not explain how scientists learn about places they cannot visit.

Connections to the anchoring phenomenon and Kenya	Repeatedly links ideas to the Nairobi night sky and gives accurate Kenyan or African examples such as Mombasa tides, Lake Victoria, 1KUNS-PF or the SKA.	Links to the Nairobi night sky and gives at least one accurate Kenyan or African example.	Gives little connection to the phenomenon or Kenyan examples.
Communication and Claim-Evidence-Reasoning structure	Writes a clear, well-organised answer with a claim, evidence and reasoning, using correct science vocabulary and a clear reflection on how thinking has changed.	Writes a mostly clear answer with a claim and some evidence and reasoning, and a short reflection.	Writes an unclear or incomplete answer that lacks a claim or reasoning.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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